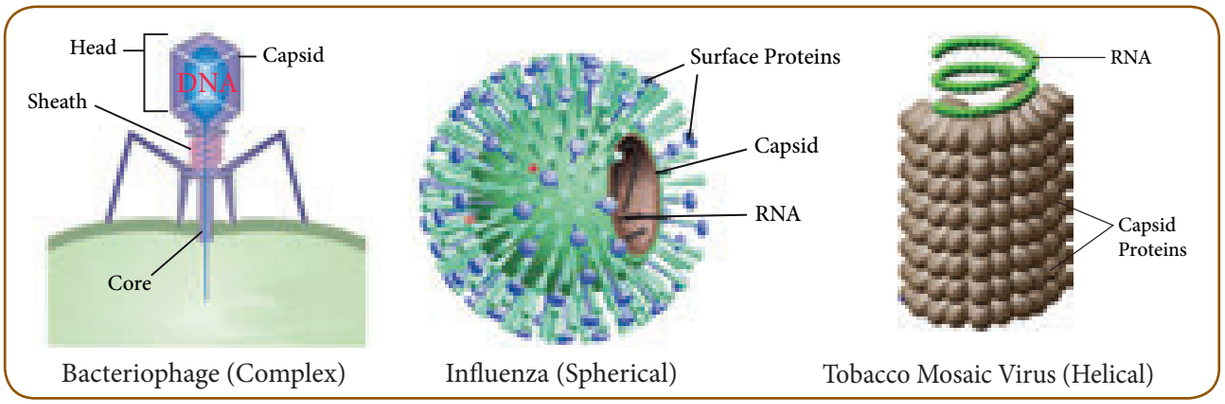


Figure 16.1 Shapes of Virus

16.2 Bacteria

Bacteria are single-celled prokaryotes (cells without nuclei). They are considered to be the first living organisms on earth. Bacteria are grouped under the kingdom Monera. The study of bacteria is called bacteriology. The size of bacteria ranges from 1µm to 5µm (micrometer). Bacteria are of two types based on respiration. They are:

- Aerobic bacteria (requires oxygen).
- Anaerobic bacteria (does not require oxygen).

16.2.1 Cell structure of Bacteria

A bacterium has an outer covering known as the cell wall. Nuclear material is represented by a nucleoid without nuclear membrane. An extra chromosomal DNA called plasmid is present in the cytoplasm. Protein synthesis is carried out by 70S ribosomes. Other cell organelles (mitochondria, golgi body, endoplasmic reticulum etc.,) are absent. Flagella aids in locomotion.

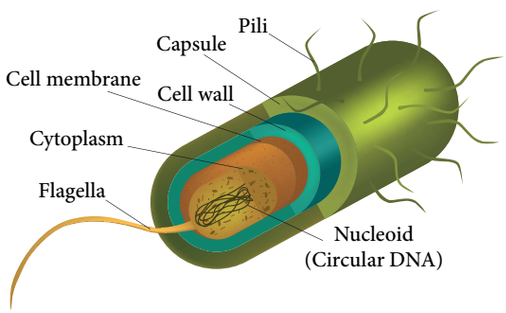


Figure 16.2 Cell structure of Bacteria

Bacteria are classified according to the shape of their cells. They are:

- Bacilli: Rod shaped bacteria.
  Eg. *Bacillus anthracis*
- Spirilla: Spiral shaped bacteria.
  Eg. *Helicobacter pylori*
- Cocci: Spherical or ball shaped bacteria. They can stick together in pairs (diplococcus) or form a chain (streptococcus) or occur in bunches (staphylococcus).
- Vibrio - Comma shaped bacteria.
  Eg. *Vibrio cholera*.

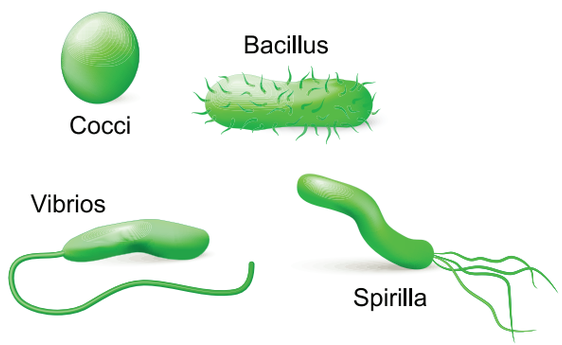


Figure 16.3 Shapes of Bacteria

Bacteria are also classified according to the number and arrangement of flagella. They are as follows.

- Monotrichous: Single flagella at one end.
  Eg. *Vibrio cholera*
- Lophotrichous: Tuft of flagella at one end.
  Eg. *Pseudomonas*.
- Amphitrichous: Tuft of flagella at both ends.
  Eg. *Rhodospirillum rubrum*.

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